

Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs (ACJC – YJC)

1. [AJC/H2/Prelims 2018/P1/4]

The given diagram shows the momentum of two trolleys, X and Y just before they collide. The collision reverses the direction of motion of both trolleys. Just after the collision, the momentum of Y is 12 N s.

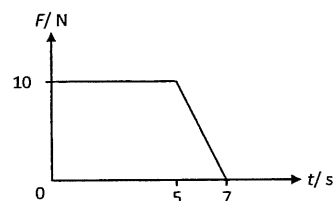


What is the magnitude of the corresponding momentum of X?

- A 4 N s B 6 N s C 10 N s D 20 N s

2. [CJC/H2/Prelims 2018/P1/5]

A body of mass 2.0 kg is moving along a smooth horizontal surface to the right with a constant velocity of 18 m s^{-1} when a force towards the left is applied for 7 s. The graph below shows how the force F varies with time t .



What is the final velocity of the body after 7 s?

- A 12 m s^{-1} to the right B 48 m s^{-1} to the right
 C 12 m s^{-1} to the left D 48 m s^{-1} to the left

3. [CJC/H2/Prelims 2018/P1/6 Modified]

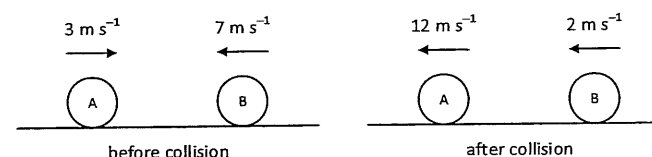
A fast-moving neutron with an initial velocity u has a head-on elastic collision with a stationary proton. After the collision, the velocity of the neutron is v and that of the proton is w .

Taking the masses of the neutron and proton to be equal, which of the following statements is incorrect?

- A Conservation of momentum shows that $u = v + w$.
 B The proton and the neutron move off in opposite directions with equal speeds.
 C By considering kinetic energies of the particles, it can be shown that $u^2 = v^2 + w^2$.
 D The speed of the proton after the collision is the same as that of the neutron before the collision.

4. [EJC/H2/Prelims 2018/P1/5]

Two objects A and B collide head-on on a horizontal, frictionless surface. The velocities of the objects before the collision and after the collision are shown in the diagram below.

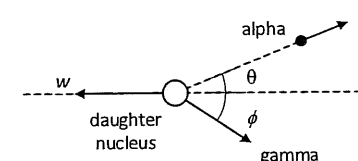


Which of the following statements is correct?

- A The sum of the kinetic energies of A and B is conserved.
 B The collision is possibly inelastic.
 C The total momentum of the two objects cannot be conserved since no net external force acts on them.
 D The momentum of each object is conserved.

5. [HJC/H2/Prelims 2018/P1/6]

A stationary nucleus of mass number A undergoes a radioactive decay by emitting an alpha particle with velocity v and a gamma radiation of wavelength λ . The daughter nucleus moves off with velocity w .



Which of the following equations is correct?

- A $(A-4)w = 4v \cos \theta$
 B $(A-4)uw = 4uv \cos \theta + \frac{hc}{\lambda} \cos \phi$
 C $4uv \sin \theta = \frac{hc}{\lambda} \sin \phi$
 D $4uv \sin \theta = \frac{h}{\lambda} \sin \phi$

6. [IJC/H2/Prelims 2018/P1/5]

Two equal masses X and Y are moving towards each other on a frictionless air track as shown. The masses make an elastic collision.

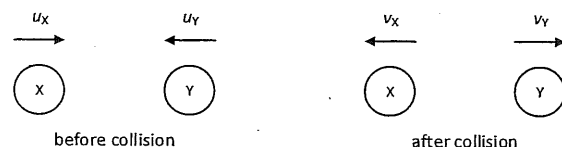


Which row gives possible velocities for the two masses after the collision?

	Velocity of X	Velocity of Y
A	zero	20 cm s ⁻¹ to the right
B	10 cm s ⁻¹ to the right	10 cm s ⁻¹ to the right
C	20 cm s ⁻¹ to the left	zero
D	30 cm s ⁻¹ to the left	50 cm s ⁻¹ to the right

7. [IJC/H2/Prelims 2018/P1/5]

Two balls X and Y approach each other along the same straight line and collide elastically. Their speeds are u_x and u_y respectively. After the collision they move apart with speeds v_x and v_y respectively. Their directions are shown below. Two objects A and B collide head-on on a horizontal, frictionless surface. The velocities of the objects before the collision and after the collision are shown in the diagram below.



Which of the following equations is correct?

- A $u_x - u_y = v_y - v_x$
 B $u_x + u_y = v_x - v_y$
 C $u_x + u_y = v_x + v_y$
 D $u_x - u_y = v_x - v_y$

8. [MI/H2/Prelims 2018/P1/6]

A missile of mass $3M$ was projected with a velocity of 50 m s^{-1} at an angle of 80° above the horizontal. At the highest point, it explodes into three equal fragments X, Y and Z. Fragment X moves vertically upwards with velocity of 25 m s^{-1} . Fragment Y moves vertically downwards with velocity of 25 m s^{-1} .

What is the velocity of fragment Z?

- A 8.6 m s^{-1} horizontally.
 B 26 m s^{-1} horizontally.
 C 8.6 m s^{-1} at an angle of 80° above the horizontal.
 D 26 m s^{-1} at an angle of 80° above the horizontal.

9. [MJC/H2/Prelims 2018/P1/5]

A body P of mass 2.0 kg and moving with velocity 3.0 m s^{-1} makes a head-on inelastic collision with a stationary body Q of mass 4.0 kg .

Which of the following could be the velocities of P and Q after the collision?

	velocity of P after collision	velocity of Q after collision
A	$+0.5 \text{ m s}^{-1}$	$+0.5 \text{ m s}^{-1}$
B	$+0.0 \text{ m s}^{-1}$	$+3.0 \text{ m s}^{-1}$
C	-1.0 m s^{-1}	$+2.0 \text{ m s}^{-1}$
D	-0.6 m s^{-1}	$+1.8 \text{ m s}^{-1}$

10. [NJC/H2/Prelims 2018/P1/4]

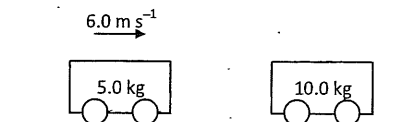
In an ice-hockey match, two players skated towards each other. After colliding head-on, only one of them was thrown backward.

The player was thrown backwards because he

- A exerted a smaller force on the other player.
 B had a smaller initial momentum.
 C had a lower initial speed.
 D had a smaller mass.

11. [NYJC/H2/Prelims 2018/P1/5]

A trolley of mass 5.0 kg travelling at a speed of 6.0 m s^{-1} collides head-on and locks together with another trolley of mass 10.0 kg which is initially at rest. The collision lasts for 0.20 s .

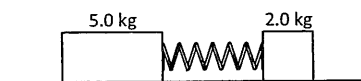


What is the total kinetic energy of the two trolleys after the collision and the average force acting on each trolley during this collision?

	Total kinetic energy after the collision / J	Average force on each trolley / N
A	30	150
B	75	150
C	30	100
D	75	100

12. [PJ/C/H2/Prelims 2018/P1/3]

A light spring is connected between two blocks of wood of masses 5.0 kg and 2.0 kg, on a frictionless horizontal surface. The spring is compressed and the blocks are released simultaneously from rest.

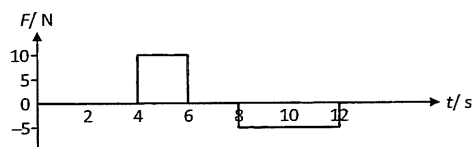


When the acceleration of the heavier block is 10 m s^{-2} , what is the acceleration of the lighter block?

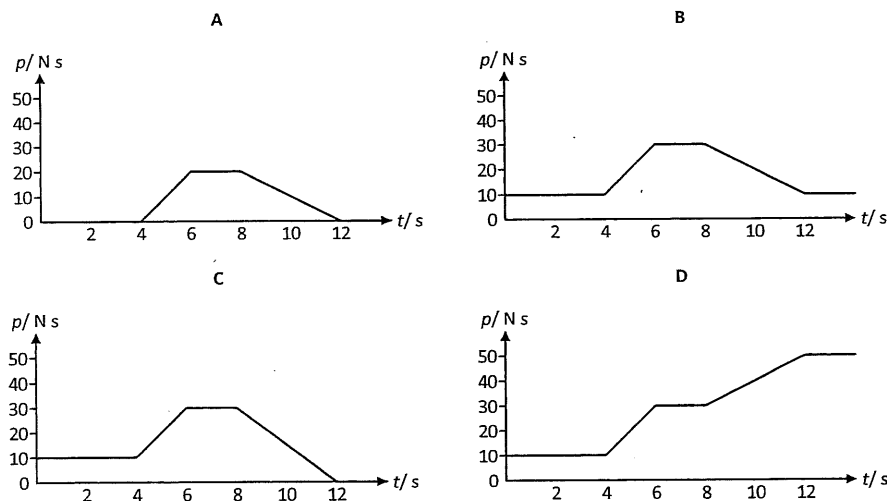
- A 5.0 m s^{-2} B 10 m s^{-2} C 20 m s^{-2} D 25 m s^{-2}

13. [TPJC/H2/Prelims 2018/P1/9 Modified]

A body of mass 2.0 kg is moving at a velocity of 5.0 m s^{-1} at time $t = 0 \text{ s}$. The resultant force F on the body varies with time t as shown.

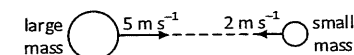


Which one of the following graphs best represents how the momentum p of the object varies with time t ?



14. [TPJC/H2/Prelims 2018/P1/10]

A large mass moving at a velocity of 5 m s^{-1} collides head-on with a small mass moving at a velocity of 2 m s^{-1} in the opposite direction.



The collision is elastic. After the collision, both masses move to the right. The large mass has a velocity v_1 and the small mass has a velocity v_2 .

Which pair of values v_1 and v_2 is possible?

	v_1	v_2
A	2 m s^{-1}	5 m s^{-1}
B	3 m s^{-1}	10 m s^{-1}
C	4 m s^{-1}	4 m s^{-1}
D	5 m s^{-1}	12 m s^{-1}

15. [YJC/H2/Prelims 2018/P1/3]

A ball of mass 0.050 kg, initially moving at 0.40 m s^{-1} , is struck by a racket exerting an average force of 250 N in the opposite direction for 1.2 ms.

What is the final speed of the ball?

- A 5.6 m s^{-1} B 6.0 m s^{-1} C 6.4 m s^{-1} D 12.2 m s^{-1}

16. [YJC/H2/Prelims 2018/P1/4]

A sub-atomic particle of mass 0.113 u collides head-on elastically with a stationary neutron.

If the neutron moves off with a speed of $3.80 \times 10^6 \text{ m s}^{-1}$, what is the initial speed of the sub-atomic particle?

- A $1.49 \times 10^7 \text{ m s}^{-1}$ B $1.87 \times 10^7 \text{ m s}^{-1}$ C $3.74 \times 10^7 \text{ m s}^{-1}$ D $4.85 \times 10^7 \text{ m s}^{-1}$

II. Structured Questions (ACJC – VJC)

1. [ACJC/H2/Prelims 2018/P3/1]

- a. A soldier fires his semi-automatic rifle in a stationary position as shown in Fig. 1.1. Each bullet has a mass of 15 g and it leaves the gun with a velocity of 700 m s^{-1} .



Fig. 1.1

- i. If the rifle weighs 4.5 kg, calculate the velocity of the rifle when he fires a single shot.

[2]

- ii. If the soldier now fires 10 bullets horizontally in 2.0 s, calculate the average force exerted by the bullets on the rifle. [3]
- b. Two balls X and Y having the same mass of 160 g collide head-on with each other as shown in Fig. 1.2.

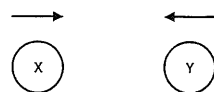


Fig. 1.2

The graph in Fig. 1.3 shows the momentum of the balls before, during and after the collision.

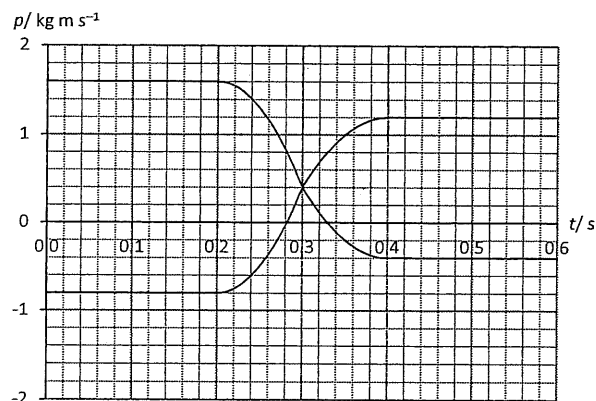


Fig. 1.3

- i. Show that the collision is inelastic. [3]
- ii. Calculate the impulse experienced by ball X due to the collision. [2]
- iii. Explain why both balls cannot lose all their kinetic energy at the same time during the collision. [2]
- c. Block B of mass 8.0 kg rests on a smooth 30° incline and is connected to block A of mass 12.0 kg via a frictionless pulley using an inextensible string as shown in Fig. 1.4. The blocks are released from rest and come to rest again momentarily when the spring is compressed by 1.2 m. The spring has a spring constant k .

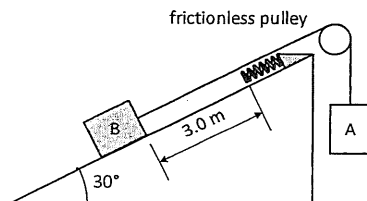


Fig. 1.4

- i. Describe the energy transformation from the time the blocks are released to the time when the blocks first come to rest. [3]
- ii. Calculate the value of k . [2]

[Total: 17]

2. [CJC/H2/Prelims 2018/P3/7 Part Modified]

- b. i. State the principle of the conservation of momentum. [1]
- ii. Body A of mass 1.2 kg collides head-on and elastically with body B of mass 0.60 kg moving with a speed 0.20 m s^{-1} towards it as shown in Fig. 2.1.

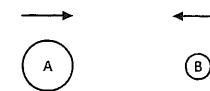


Fig. 2.1

After the collision, object B moves off with a speed of 0.10 m s^{-1} opposite to its initial motion.

Calculate the initial speed of object A. [3]

- c. An 80 kg astronaut is at a distance of 30 m away from a space shuttle. He wishes to return to the space shuttle by means of a thruster. The thruster is attached to the body of the astronaut and it emits a stream of gas when it is turned on.
- i. State and explain the direction in which the gas has to be ejected for him to return to the space shuttle. [3]
- ii. The gas is ejected at a constant rate of 0.50 kg s^{-1} and at a speed of 20 m s^{-1} relative to the astronaut for a period of 1.0 s. Calculate the speed of the astronaut at the end of 1.0 s. [2]

[Total: 9]

3. [HCI/H2/Prelims 2018/P2/2]

- a. State the principle of conservation of linear momentum. [1]
- b. A light and long string (string 1) runs over two smooth and light pulleys (pulley A and B). One end of the string is fixed to the ceiling and the other end is attached to mass m_1 . Pulley A is fixed to the ceiling while pulley B is movable and is attached to mass m_2 via another light string (string 2).

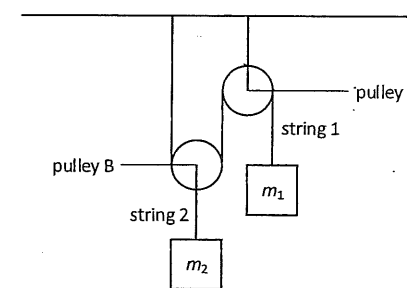


Fig. 3.1

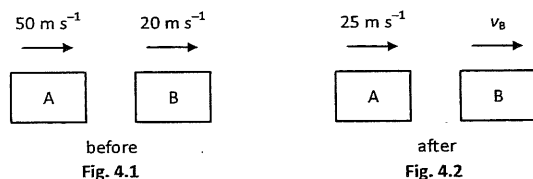
- i. Given that $m_1 = 4.0 \text{ kg}$ and $m_2 = 8.0 \text{ kg}$ and they are both at rest, determine the tension in string 1 and string 2. [2]
- ii. A little disturbance is made to the system and m_1 starts to move upward with speed v_1 while m_2 starts to move downwards with v_2 . By considering linear momentum, express v_2 in terms of v_1 . [2]
- iii. Hence, determine the energy introduced to the system due to the disturbance in terms of v_1 . [2]
- iv. After some time, the masses move with constant speed. Show that total energy of the system is conserved. Explain your working clearly. [2]

[Total: 9]

4. [JJC/H2/Prelims 2018/P2/3]

A 900 kg car A travelling at 50 m s^{-1} collides inelastically with a 1500 kg car B travelling at 20 m s^{-1} in the same direction as shown in Fig. 4.1. Immediately after the collision, the velocity of car A is 25 m s^{-1} and the velocity of car B is V_B as shown in Fig. 4.2.

The collision begins at time t . During the collision, both cars are in contact for 120 ms.



- a. i. Determine the magnitude of the velocity V_B . [1]
- ii. On Fig. 4.3, sketch and label the momentum-time graphs for car A and car B, before, during and after the collision. Include numerical values.

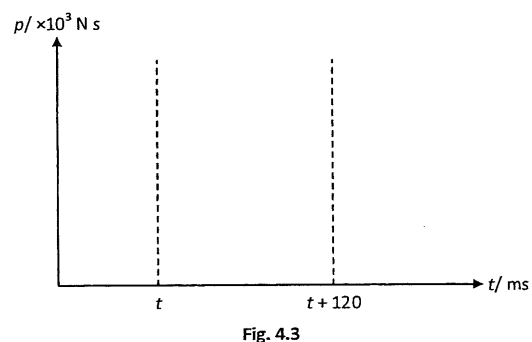


Fig. 4.3

- b. Determine the average force that car B exerts on car A during the collision. [1]
- c. State and explain one design or feature of the car that reduces the force experienced by the driver. [2]

[Total: 8]

5. [MI/H2/Prelims 2018/P3/7 Modified]

- b. Two balls, X and Y are travelling towards each other along a horizontal frictionless surface.
- i. Show how the principle of conservation of linear momentum can be derived using Newton's Laws. [3]
- ii. 1. The two balls collided and rebounded in opposite directions. Fig. 5.2 shows data for X and Y during the collision. The positive direction is horizontal and to the right.

ball	mass / g	velocity just before collision/ m s^{-1}	velocity just after collision/ m s^{-1}
X	50	+4.5	-1.8
Y	M	-2.8	+1.4

Fig. 5.2

Use your answer in b.i. to determine the mass of Y. [3]

2. State and explain quantitatively whether the collision is elastic. [1]
- iii. The variation of the force that X exerts on Y with time is shown in Fig. 5.3.

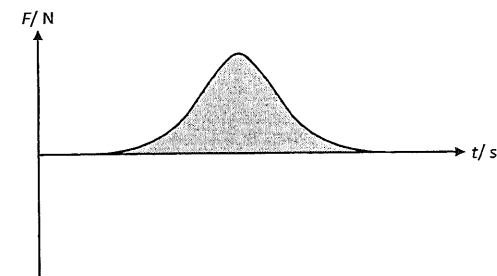


Fig. 5.3

1. Sketch, on Fig. 5.3, the variation of the force that Y exerts on X with time. Explain your answer. [2]
2. Explain what the shaded area in Fig. 5.3 represents. [1]

[Total: 10]

6. [MJC/H2/Prelims 2018/P3/3]

- a. State Newton's second law of motion. [1]
- b. A car of mass 800 kg was travelling on a horizontal road at a constant speed of 20 m s^{-1} before a net horizontal constant forward force of 4800 N acts on the car for 12 s.
- Calculate
- i. the distance travelled by the car over the 12 s. [2]
- ii. the speed of the car at the end of the 12 s. [2]
- iii. the work done on the car during the 12 s
1. using the answer to b.i. [1]
2. using the answer to b.ii. [1]
- iv. the impulse exerted on the car over the 12 s. [2]

[Total: 9]

7. [MJC/H2/Prelims 2018/P3/5]

Ball A, of mass 800 g and travelling with a speed of 9.2 m s^{-1} , collided head-on with a stationary ball B of mass 2400 g. The collision is completely inelastic.

- Explain whether the total momentum is conserved during the collision. [1]
- Calculate the percentage loss in total kinetic energy. [2]
- Shortly after the collision, Ball B comes into contact with a spring of spring constant 2500 N m^{-1} . Calculate the maximum compression of the spring. [2]

[Total: 5]

8. [NJC/H2/Prelims 2018/P2/2 Modified]

- By reference to Newton's laws of motion, explain why when two particles, which are isolated, collide, the total momentum of the particles is conserved. [3]
- A ferryboat of mass $M_1 = 7.0 \times 10^5 \text{ kg}$ moves toward a docking bumper of mass M_2 that is attached to a shock absorber as shown in Fig. 8.1.

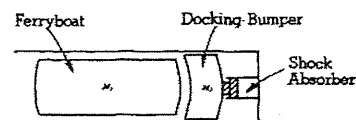


Fig. 8.1

Just before hitting the bumper, the ferryboat is travelling at a speed $v_i = 3.0 \text{ m s}^{-1}$. After colliding inelastically with the bumper, the ferryboat and bumper move together with an initial speed of 2.0 m s^{-1} . The ferryboat comes to a complete stop 4.0 s after the collision. Fig. 8.2 shows the variation with time t of the velocity v of the ferryboat, where $t = 0 \text{ s}$ is the instant of collision.

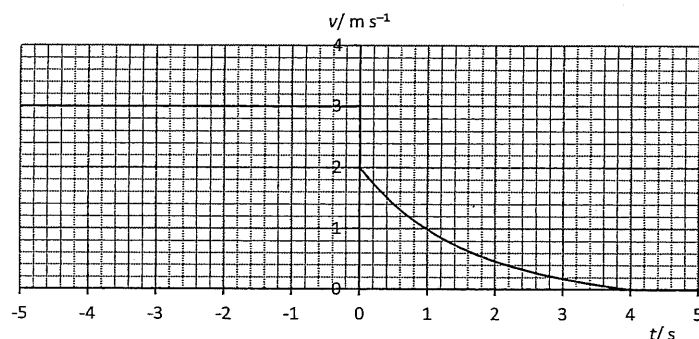


Fig. 8.2

- Explain what is meant by an *inelastic collision*. [1]
- Calculate the mass of the bumper M_2 . [2]
State an important assumption that you have made in the calculation.
- Use the graph to determine the magnitude of the acceleration of the ferryboat at $t = 1.0 \text{ s}$. [3]
Show your working on the graph clearly. State the direction of the acceleration.
- Hence calculate the magnitude of the retarding force on the ferryboat at $t = 1.0 \text{ s}$. [1]

[Total: 10]

9. [NJC/H2/Prelims 2018/P3/2 Modified]

An executive toy consists of five identical steel spheres of mass m suspended so that they are free to move in a vertical plane as shown in Fig. 9.1. Each sphere is suspended using thin inextensible strings of negligible mass. You may assume that collisions between the spheres are perfectly elastic and air resistance is negligible.

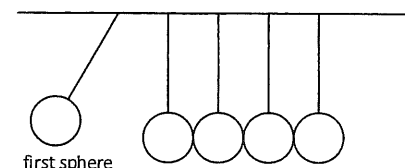


Fig. 9.1

- The first sphere is displaced to the left and then released. Describe and explain the resulting motion of the five spheres. [5]
- Object A has a velocity of u and moves horizontally towards object B which is at rest. The collision between both objects is head-on and perfectly elastic. The ratio of the mass of object A to the mass of object B is 1:2. Determine the ratio of the final velocity of object B to the initial velocity of object A. [2]
 - Object C has a velocity of u and moves horizontally towards Object D which is at rest. The collision between both objects is head-on and perfectly elastic. The ratio of the mass of object C to the mass of object D is 2:1. Determine the ratio of the final velocity of object D to the initial velocity of object C. [1]
 - Hence, if the second and fourth spheres are replaced with steel spheres of mass $2m$ of the executive toy in Fig. 9.1, determine the ratio of

$$\frac{\text{maximum height of first sphere}}{\text{maximum height of fifth sphere}}$$

Assume that the collision between both objects is head-on and perfectly elastic. [2]

- The five identical steel spheres in Fig. 9.1 are of mass m . State and explain one significant difference to the motion observed if all the thin inextensible strings are replaced by springs of negligible mass. [2]

[Total: 12]

10. [NYJC/H2/Prelims 2018/P2/1 Modified]

In 2018, the Singapore government announced that personal mobility devices (PMDs) such as e-scooters used on public paths must not have a speed exceeding 25 km h^{-1} or a mass of more than 20 kg.

- Suggest why there is a need to set a speed limit and mass limit on PMDs. [2]
- Given that a typical e-scooter with rider has a total mass of 100 kg and that the time of impact is 0.20 s, determine the average force this scooter could cause. Assume that the e-scooter comes to rest after impact. [2]
- Other than the speed and mass of PMDs, state and explain one other factor that can contribute to the magnitude of the force of impact caused in an accident involving PMDs. [2]
- It is suggested that an infra-red sensor can be fixed at the front end of a PMD such that once an obstacle is detected within 30 cm, the PMD will decelerate to a stop. By reference to the maximum allowable speed of 25 km h^{-1} , explain whether this is feasible. [2]

[Total: 8]

11. [SRJC/H2/Prelims 2018/P3/1]

- a. Define *impulse*. [1]
- b. In a car test, a car with a dummy driver and passenger, moving at a speed of 6.9 m s^{-1} , collides head-on into a wall. The mass of the car is 1250 kg, the mass of the driver is 85 kg and the mass of the front passenger is 65 kg. The average deceleration of the car as it comes to a stop is 48 m s^{-2} . Both passenger and driver have their seat belts tightly fastened. For the impact,
- determine the magnitude of the average force exerted on the car and its occupants, [1]
 - determine the magnitude of the impulse caused by the force. [2]
 - Hence, calculate the time taken for the car to come to a stop. [2]
 - Assuming that the average deceleration remains the same, state and explain how your answer in b.iii. will change (if any) when the total mass of car and occupants is doubled. [2]

[Total: 8]

12. [VJC/H2/Prelims 2018/P3/2]

- a. i. A man pushes a heavy rock resting on the ground, but it does not move. A student says that this is because the pushing force is balanced by the reaction to this force. Explain why the student's argument is wrong. [2]
- ii. Draw a labelled diagram to show the forces acting on the rock. [2]
- b. Block A of mass 2.0 kg moves with a velocity 10 m s^{-1} to the right on a smooth horizontal table. Block B of mass 3.0 kg moves with a velocity of 5.0 m s^{-1} in front of A in the same direction. A light spring of force constant 500 N m^{-1} is attached to B as shown in Fig. 12.1.

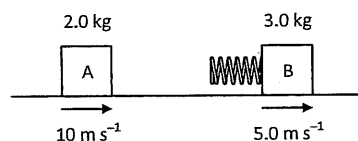


Fig. 12.1

- When A collides with B, there will be an instant when the spring experiences maximum compression. Describe the motion of A and B when maximum compression of the spring takes place. [1]
- Calculate the final velocity of the two blocks at the instant the spring is at its maximum compression. [2]
- Hence calculate the maximum compression of the spring. [2]

[Total: 9]

13. [PJC/H2/Prelims 2018/P3/2]

A rigid bar of mass 450 g is held horizontally by two supports A and B, as shown in Fig. 13.1.

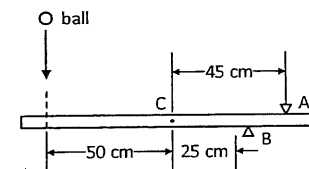


Fig. 13.1

The support A is 45 cm from the centre of gravity C of the bar and support B is 25 cm from C. A ball of mass 140 g falls vertically onto the bar such that it hits the bar at a distance of 50 cm from C, as shown in Fig. 13.1.

The variation with time t of the velocity v of the ball before, during and after hitting the bar is shown in Fig. 13.2.

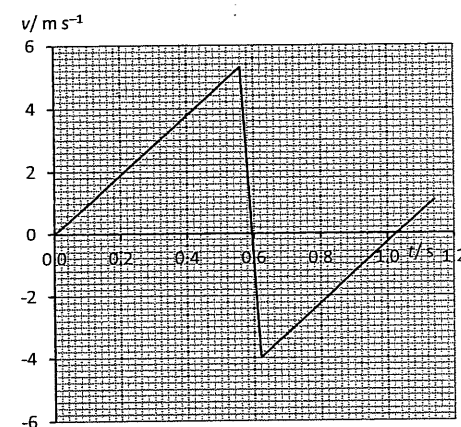


Fig. 13.2

- For the time that the ball is in contact with the bar, use Fig. 13.2 to determine
 - the change in momentum of the ball, [2]
 - the magnitude of the force exerted by the ball on the bar. [3]
- For the time that the ball is in contact with the bar, use data from Fig. 13.1 and a.ii. to calculate the magnitude of the force exerted on the bar by
 - support A, [2]
 - support B. [1]

[Total: 8]

14. [TPJC/H2/Prelims 2018/P3/2]

- a. A horizontal force of 54 N is applied to push two touching blocks of mass 6.0 kg and 2.0 kg along a flat surface. The frictional force between the blocks and the surface is 6.0 N.

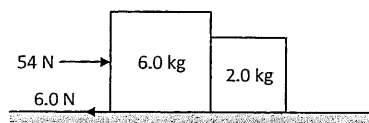


Fig. 14.1

Calculate the magnitude of the resultant force on the 6.0 kg mass.

[3]

- b. A metal block X of mass 0.015 kg is dropped from rest from a certain height until it hits a table Y.
- i. Using the diagram below, for the instant of impact between the metal block X and table Y, draw and label all forces acting on X and Y. Make it clear which forces are equal in magnitude and opposite in direction.

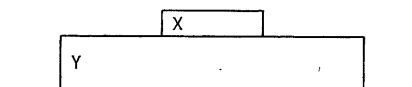


Fig. 14.2

[3]

- ii. Just before hitting the table-top, the metal block has a speed of 3.92 m s^{-1} . Upon impact with the table-top, the metal block remains in contact with the table-top for a duration of 1.5 ms before it bounces vertically off the table-top with a 20% reduction in speed.
1. Calculate the change in momentum of the metal block after its collision with the table-top. [2]
 2. Hence determine the average force exerted by the metal block on the table-top. [2]

[Total: 10]